

In-Class Exercise Sheet 11

Questions 1–5 are based on the seminal article Nature “The FAIR Guiding Principles for scientific data management and stewardship” by M. D. Wilkinson et al. (2016). Question 7 and onward are based on the articles “Legal Perspectives on Research Data Storage” and “Datenbankherstellerrecht und Datenbankforschung” by M. Beurskens and S. Scherzinger, published in the journal *Datenbank-Spektrum* (2023 and 2024 resp.).

1. The FAIR Guiding Principles apply to which of the following research artefacts? Select all that apply.

- ☐ Research data
- ☐ Algorithms used to process or analyze the data
- ☐ Software tools used in the research process
- ☐ Workflows that led to the data
- ☐ Physical laboratory equipment

2. Which of the following are considered stakeholders? Select all that apply.

- ☐ The researchers who conducted the original work
- ☐ Researchers who want to reuse each other’s data
- ☐ Professional data publishers
- ☐ Funding agencies
- ☐ Computational agents that discover and process data

3. A central goal of the FAIR Guiding Principles is that digital objects should be

_____.

enabling _____ to autonomously discover, interpret, assess, and reuse data.

4. Which strategy best supports good data stewardship?

- ☐ Create bespoke parsers for every new data type.
- ☐ Adopt general-purpose, open interoperability standards.
- ☐ Restrict repositories to a small set of approved data types.
- ☐ Require every analysis tool to support a wide range of data formats.

5. Complete the four FAIR categories and review their guiding principles.

(a) To be _____:

- (Meta)data are assigned a globally unique and persistent identifier.
- Data are described with rich metadata.
- Metadata clearly and explicitly include the identifier of the data they describe.
- (Meta)data are registered or indexed in a searchable resource.

(b) To be _____:

- (Meta)data are retrievable by their identifier using a standardized communications protocol.
- The protocol is open, free, and universally implementable.
- The protocol allows for authentication and authorization where necessary.
- Metadata remain accessible even when the data are no longer available.

(c) To be _____:

- (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- (Meta)data use vocabularies that themselves follow the FAIR principles.
- (Meta)data include qualified references to other (meta)data.

(d) To be _____:

- (Meta)data are richly described with accurate and relevant attributes.
- (Meta)data are released with a clear and accessible data usage license.
- (Meta)data are associated with detailed provenance.
- (Meta)data meet domain-relevant community standards.

6. Assign each statement to one of the four FAIR categories.

(a) The (meta)data are released with a clear and accessible data usage license.

☐ Findable ☐ Accessible ☐ Interoperable ☐ Reusable

(b) The (meta)data are assigned a globally unique and persistent identifier.

☐ Findable ☐ Accessible ☐ Interoperable ☐ Reusable

(c) The research data and metadata use commonly adopted, accessible, and preferably open standards and formats.

☐ Findable ☐ Accessible ☐ Interoperable ☐ Reusable

(d) Metadata remain accessible even when the data are no longer available.

☐ Findable ☐ Accessible ☐ Interoperable ☐ Reusable

Exercise translated from the textbook “K. Biernacka and S. Schulz: “Forschungsdatenmanagement in der Informatik”, Logos (2022). Original source: Asef, E., Bierndicka, K., Böker, E., Danker, S. A., Juliane, J., Neumann, J., Petersen, B., Rex, J., & Trautwein-Bruns, U. (2021). “Data Sharing interaktiv vermitteln”. <https://doi.org/10.5281/zenodo.4585487>.

7. Match each legal framework with the scenario to which it most directly applies.

- | | |
|----------------------------|---|
| a) Data protection (GDPR) | b) A company keeps its customer list confidential using NDAs and access restrictions. |
| c) Copyright | d) A spreadsheet contains personal information about identifiable living individuals. |
| e) Trade secret protection | f) A research article reproduces an original figure published in a textbook. |
| g) None of the above | h) A database contains only publicly available weather measurements. |

8. Which legal framework most directly protects the following database artifacts?

- (a) A database schema, provided that it is sufficiently original.
☐ Data protection ☐ Copyright ☐ Trade secret protection
- (b) A complex stored procedure or user-defined function developed for a database system.
☐ Data protection ☐ Copyright ☐ Trade secret protection
- (c) An original graphical user interface (GUI) for accessing a database.
☐ Data protection ☐ Copyright ☐ Trade secret protection
- (d) An original web API for interacting with a database.
☐ Data protection ☐ Copyright ☐ Trade secret protection
- (e) An image of Bart Simpson, stored as BLOB inside a database table.
☐ Data protection ☐ Copyright ☐ Trade secret protection

9. For each of the following scenarios, indicate whether the database schema is likely to be protected by copyright.

- (a) A database stores people’s names using two attributes, `firstName` and `lastName`.
☐ Yes ☐ No
- (b) A pharmaceutical company develops a novel database schema that partitions Germany into 1,860 carefully crafted geographical regions for standardized analysis of regional drug sales.
☐ Yes ☐ No

10. In the European Union, a specific right may apply to a database instance as such.

This right is called _____.

It requires a “qualitatively and/or quantitatively a substantial investment in either the obtaining, verification or presentation of the content” and prohibits “extraction and/or re-utilization of the whole or of a substantial part of the contents of that database”.

The following questions focus on this specific right.

- (a) The British Horseracing Board (BHB) organized horse races and maintained a database containing information about races and participating horses.

Creating this information required substantial effort and expense!

The BHB maintained a massive database of racing information, such as lists of runners, riders, and race dates. The bookmaker William Hill used this data on its betting website. The BHB then sued William Hill.

How do you think the European Court of Justice ruled with respect to this database?

- ☐ The database was protected.
- ☐ The database was not protected.

- (b) The German Toll Collect GmbH collects and maintains records generated by the German truck toll system. These records are supplied to a payment-service provider, which reuses the data for its customers.

Obtaining this information requires substantial effort and expense!

How do you think the German Federal Court of Justice (*Bundesgerichtshof*, BGH) ruled with respect to this database?

- ☐ The database was protected.
- ☐ The database was not protected.

- (c) A university finances the creation of a research database by employing several researchers and providing the necessary infrastructure. Who owns the rights to the database?

- ☐ All researchers jointly, because they created the database.
- ☐ The university, because it made the substantial investment.
- ☐ The principal investigator who supervised the project.
- ☐ None of the above.

- (d) Which of the following actions can violate this right? Select all that apply.

- ☐ Extracting a single record from a database.
- ☐ Repeatedly extracting individual records until the database has effectively been copied.
- ☐ Extracting all records from the database.

- (e) A US company compiles a large database and puts it on their website. It does not require users to accept a license agreement before accessing it.

Can the US company claim their database is protected? ☐ Yes ☐ No

- (f) A German research institution publishes a database on its website without requiring users to accept a license agreement before accessing it. A company based in the United States copies the database and uses it to develop an innovative product.

Can the research institute claim their database is protected? ☐ Yes ☐ No

- (g) A German research institution publishes a database on its website without requiring users to accept a license agreement before accessing it. A research team from France would like to reproduce the experiments and compare their data against the database.

Does the French team risk violating the sui generis database right? ☐ Yes ☐ No

11. Label the overview chart below, assigning the following terms.

Privacy / IP Rights / Trade Secrecy / Copyright in Software / Consent or License / Copyright in Structure / sui generis rights / Terms of Use / License / Technical Framework / Individuals / Support / Institutions / Contract Terms / Privileged Use / Restricted Use / Attribution / Cross-Cutting Concerns: Security, Compliance, Availability

